

Olist E-Commerce Marketplace Performance & Customer Experience Analysis

Analysis Report | Evidence-based operational insights

1. Problem Understanding

Olist operates as a marketplace integrator connecting Brazilian small and medium merchants with major marketplaces. The objective is to understand how marketplace growth, delivery performance, seller behavior, product categories, geography and payment behavior interact with customer satisfaction, and to identify operational priorities supported by data.

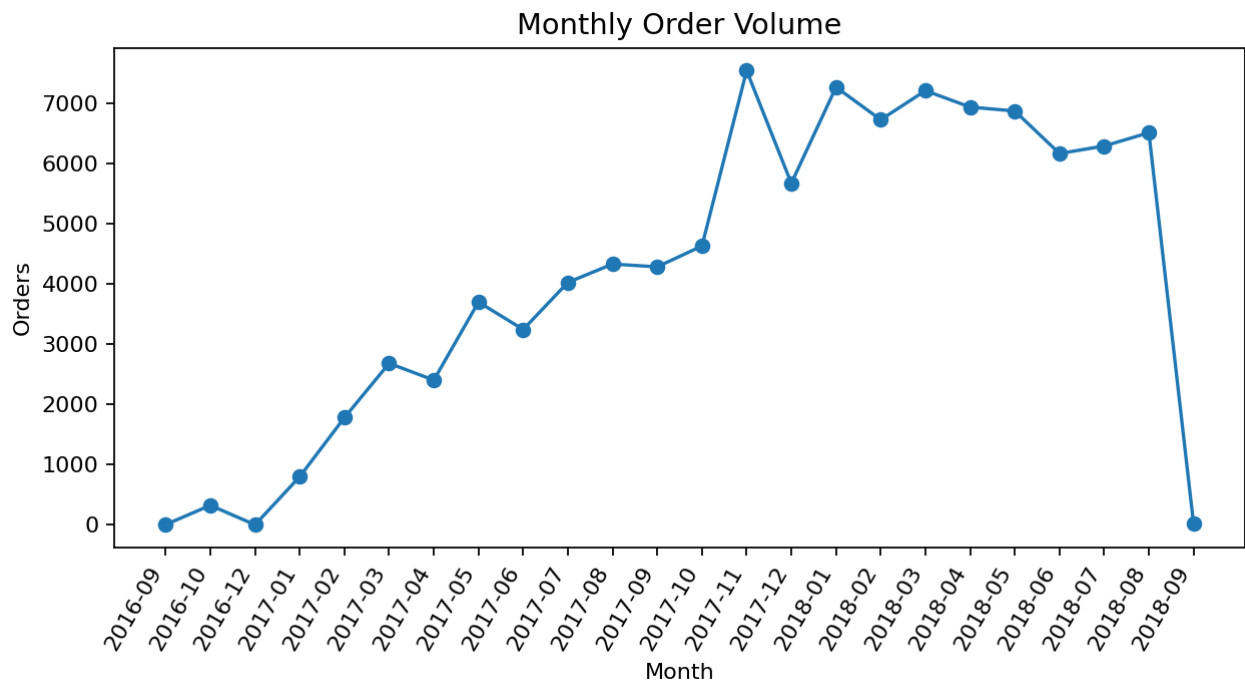
2. Analytical Approach

The analysis integrates orders, order items, payments, reviews, customers, sellers, products, geolocation and category-translation data. Different table grains were kept separate where necessary to avoid double-counting. Dates were standardized and features were created for delivery duration, delivery delay, satisfaction, freight ratio and time periods.

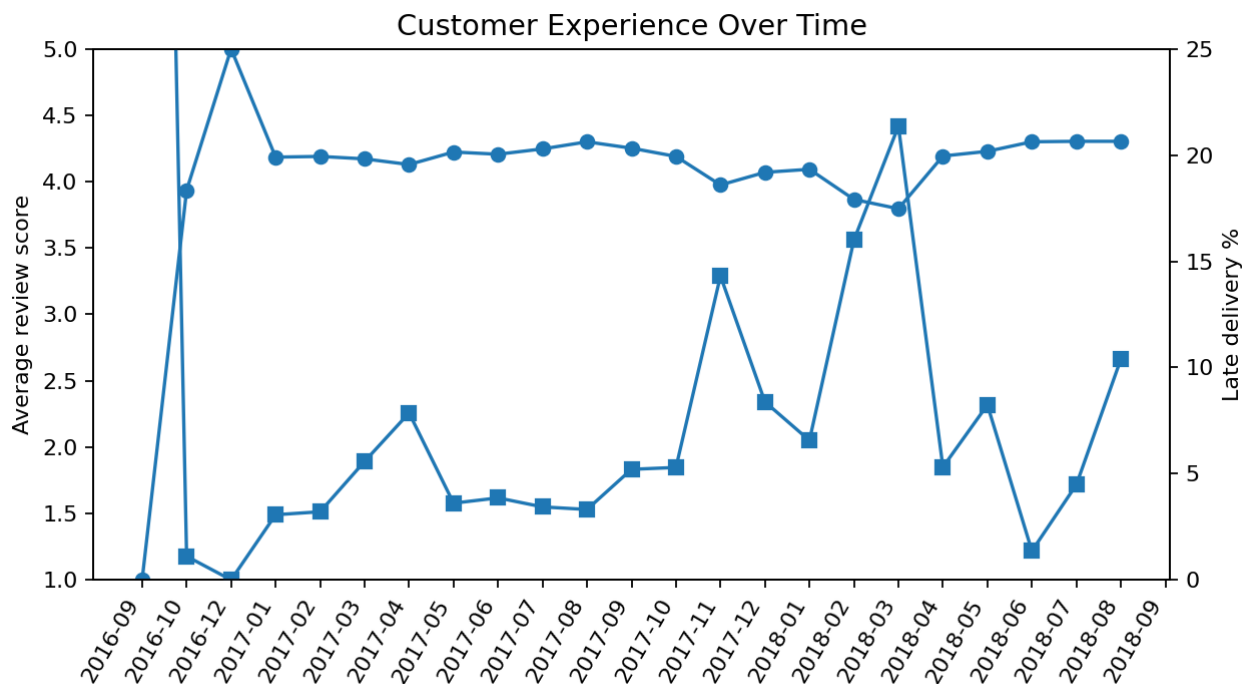
3. Executive Performance Snapshot

KPI	Result
Total orders	99,441
Unique customers	96,096
GMV / product revenue	R\$ 13,591,643.70
Freight value	R\$ 2,251,909.54
Average review score	4.07 / 5
Satisfied reviews (4-5)	76.60%
Dissatisfied reviews (1-2)	15.05%
Average delivery days	12.56
Median delivery days	10.22
Late delivery rate	8.11%
Cancellation rate	0.63%

4. Marketplace Performance Over Time

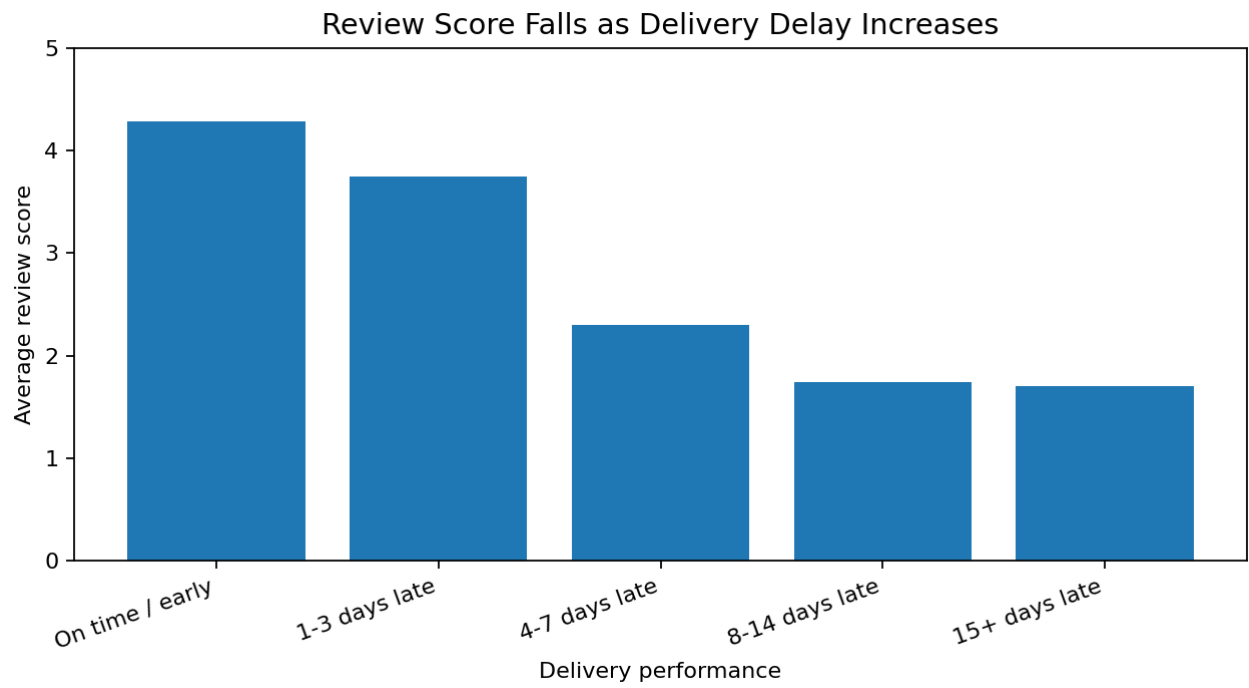


Order volume expanded strongly through 2017 and 2018. The highest monthly order count in the available series was 7,544 in November 2017. Periods of elevated late delivery coincided with weaker customer experience, making operational reliability important as the marketplace scaled.



5. Customer Satisfaction & Delivery Performance

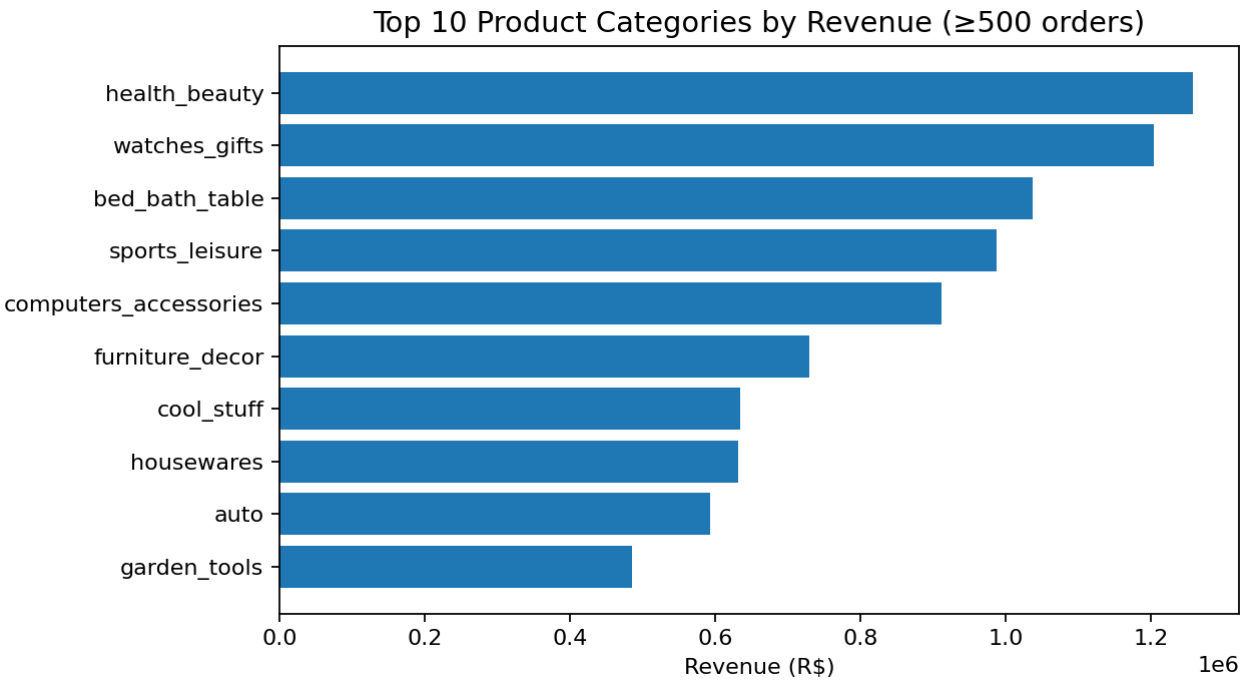
On-time/early orders average 4.28 stars versus 2.55 for late orders. Dissatisfaction is approximately 9.5% for on-time/early orders and 54.6% for late orders.



Delay severity is strongly associated with lower ratings: 4–7 day delays average about 2.30 stars, while 8–14 and 15+ day delays average below 1.75 stars and have roughly 78% dissatisfied reviews.

6. Product Category Findings

Category	Orders	Revenue	Avg. review
health_beauty	8,836	R\$ 1,258,681	4.12
watches_gifts	5,624	R\$ 1,205,006	4.00
bed_bath_table	9,417	R\$ 1,036,989	3.87
sports_leisure	7,720	R\$ 988,049	4.09
computers_accessories	6,689	R\$ 911,954	3.92



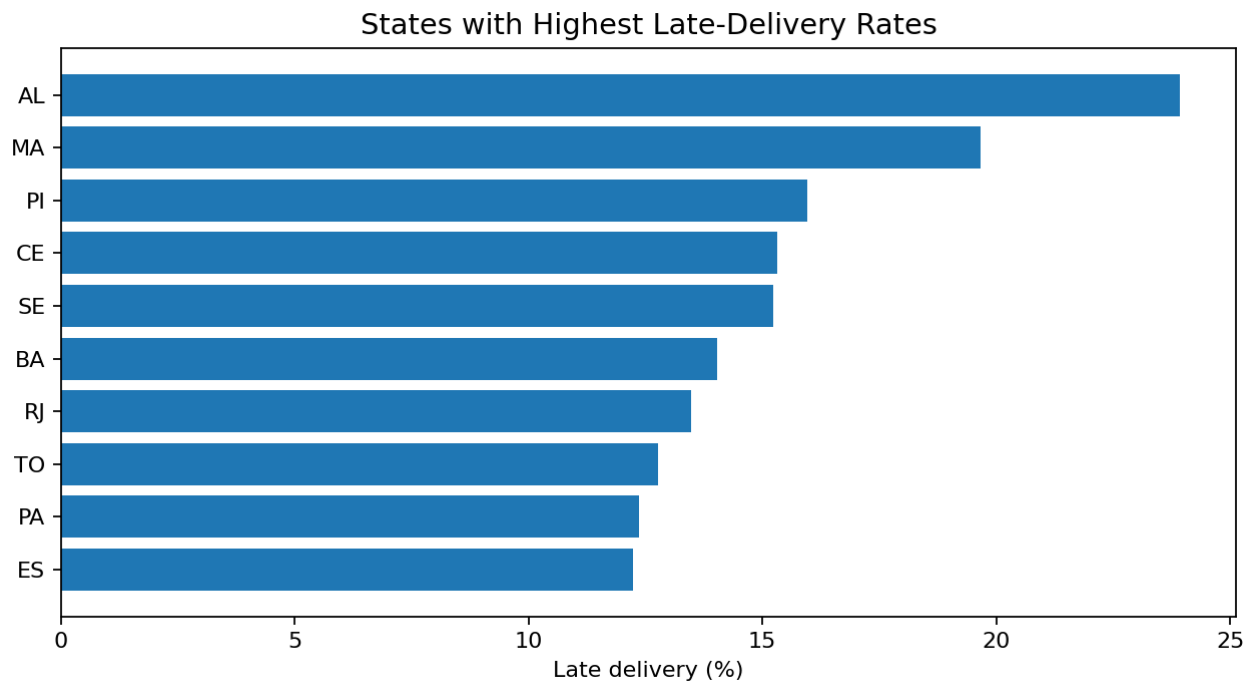
High-volume categories with weaker satisfaction are more actionable than very small categories because improvements affect more customers.

7. Seller Performance

Seller performance should be assessed using volume as well as quality. A minimum threshold of 50 orders was used for the seller deep-dive to reduce instability from small samples. High-volume sellers with elevated dissatisfaction or late delivery represent the greatest operational leverage.

Seller	Orders	Revenue	Review	Late %
1ca7077d890b...	115	R\$ 13,342	2.19	19.7%
2eb70248d66e...	202	R\$ 42,629	2.68	14.2%
54965bbe3e4f...	78	R\$ 10,961	2.83	32.1%
bbad7e518d7a...	69	R\$ 4,462	3.01	22.6%
972d0f9cf61b...	83	R\$ 8,089	2.92	11.6%
8e6d7754bc7e...	85	R\$ 14,497	2.97	12.4%
95f83f51203c...	52	R\$ 3,217	3.21	4.6%
602044f2c161...	50	R\$ 3,739	2.93	23.6%

8. Geographic Findings



São Paulo is the dominant volume market with 40,495 orders and relatively strong delivery performance. Several states, including BA, CE, MA and AL, show materially higher late-delivery rates and longer delivery times, making them candidates for route and logistics investigation.

9. Payment Behavior

Payment	Orders	Review	Dissatisfied	Installments
credit_card	76,132	4.07	15.0%	3.52
boleto	19,784	4.07	14.8%	1.00
voucher	1,994	3.94	18.9%	1.11
debit_card	1,527	4.16	13.2%	1.00
not_defined	3	1.67	66.7%	1.00

Payment type shows relatively small differences in average review compared with delivery performance. Credit card dominates volume, while voucher users show somewhat higher dissatisfaction. These are associations, not causal effects.

10. Freight Cost Deep Dive

Freight burden has a weaker but directional association with satisfaction. Average review declines from about 4.05 when freight is at most 10% of product price to about 3.94 when freight exceeds 50%; dissatisfaction rises from about 16.4% to 18.0%.

11. Key Findings

- Marketplace scale increased substantially across the observed period, but growth was accompanied by periods of elevated delivery risk.
- Delivery reliability is the strongest operational factor associated with customer satisfaction in the analysis: late orders average 2.55 stars versus 4.28 for on-time/early orders.
- Delay severity matters: once an order is 4–7 days late, average review falls to about 2.30; at 8+ days late, roughly 78% of reviews are dissatisfied.
- Performance varies geographically. High-volume SP performs relatively well, while several northern/northeastern states show much higher late-delivery rates and longer delivery times.

- Category performance is uneven. High-revenue categories with weaker review outcomes represent important opportunities because improving them can affect many customers.
- Seller performance should be managed using both scale and quality metrics; high-volume sellers with elevated late-delivery or dissatisfaction rates have the greatest operational leverage.
- Payment method has comparatively small differences in average review at the aggregate level; delivery and fulfillment should therefore receive higher priority than payment-method optimization.
- Freight burden has a weaker but measurable association with satisfaction, particularly when freight represents a large share of product price.

12. Actionable Recommendations

1. Establish a severe-delay prevention program

Use 4+ days late as an operational escalation threshold, with an additional critical threshold at 8+ days. Prioritize interventions before orders cross these thresholds.

2. Prioritize high-volume seller and route hotspots

Rank sellers and seller→customer routes by the number of late and dissatisfied orders they generate, not only by percentage metrics.

3. Create state-level logistics service targets

Set differentiated monitoring for states with high late-delivery rates and long delivery times, while protecting strong performance in high-volume markets such as SP.

4. Target high-revenue / low-satisfaction categories

Combine category revenue, order volume, review score and late-delivery rate to identify categories where operational improvements have the largest customer and commercial impact.

5. Monitor freight burden alongside delivery

Flag products/categories where freight is a high percentage of product price, particularly for lower-value purchases, and test logistics or seller-level pricing interventions.

6. Use an ongoing CX operations dashboard

Track orders, revenue, on-time delivery, severe-delay rate, 1–2 star rate, seller performance, category performance and state hotspots monthly so that deterioration is detected early.

13. Conclusion

The strongest evidence points to delivery timeliness as Olist's principal customer-experience lever. Severe delays are associated with very high dissatisfaction. Geography, seller, category and freight analysis provide additional targeting layers. Olist should focus on preventing severe delays, especially among high-volume sellers, routes and states, while monitoring category and freight hotspots.

Appendix: Analytical Scope

Source tables: orders, order items, payments, reviews, customers, sellers, products, geolocation and category translation. Review findings use orders with available review scores. Delivery-delay analysis compares delivered customer date with estimated delivery date. Seller deep-dive rankings use a minimum 50-order threshold.